Sahaj Bhadja

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Summary

Software engineer with 2+ years designing, developing, testing, and modernizing applications using Java, Python, C++, JavaScript, React, Node.js, SQL, and NoSQL. Experienced in secure integrations, visualization interfaces, cloud infrastructure, CI/CD, debugging, and documentation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Software Engineering, Database Systems, Web Technologies, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python REST APIs and Azure SQL services for 500+ users, translating product requirements into scalable features while optimizing schemas and queries to reduce response latency from 1.2 seconds to 350 milliseconds.
- Modernized a legacy workflow into maintainable services with explicit states, automated validation, and failure isolation, tripling throughput while improving code reviews, debugging, release reliability, and technical documentation.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed Flask REST services handling 10,000+ daily requests, implementing secure authentication, data workflows, validation, and error handling while troubleshooting production issues affecting more than 150 users.
- Built responsive React interfaces and multithreaded processing for 1,000+ files with relational persistence, automated tests, progress visualization, and recovery, reducing turnaround 60% under concurrent production workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built testable Python pipelines processing 100,000+ records, improving planning accuracy 18% through reusable components, automated validation, structured analysis, performance monitoring, detailed technical documentation, and systematic debugging.
- Collaborated with researchers to translate ambiguous requirements into modular software and visual analysis workflows, improving stability 15% while reviewing results and validating behavior across diverse simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed secure Python services and SQL integrations using AWS Glue, S3, filtering, pagination, and caching, improving response times 30% and processing efficiency 68% across connected enterprise applications.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito, accelerating issue investigation, regression verification, peer code reviews, and reliable CI/CD releases across multiple environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed JavaScript and Node.js services across eight microservices with authentication, REST APIs, WebSocket messaging, and state synchronization, securely processing more than two million Azure SQL database records.
- Automated builds, functional tests, staged deployments, and monitoring through Jenkins, Docker, and Azure, reducing release time 50% while improving validation coverage, release consistency, and overall operational visibility.

Skills

Languages & UI: Java, Python, C++, JavaScript, TypeScript, SQL, React, Angular.js, Next.js, Node.js, HTML, CSS, data structures, algorithms, object-oriented programming
Backend & Data: Flask, FastAPI, REST APIs, WebSockets, microservices, PostgreSQL, Azure SQL, SQL Server, MySQL, MongoDB, relational and NoSQL queries, schema design
Security & Quality: Authentication, role-based access, secure APIs, unit testing, integration testing, PyTest, Mockito, debugging, log analysis, performance monitoring, regression testing
Cloud & Delivery: AWS, Azure, S3, AWS Glue, Git, code reviews, Jenkins, GitHub Actions, CI/CD, Docker, Kubernetes, Jira, Agile, technical documentation, release processes

Projects

Chat2SQL | Code | *Java, TypeScript, SQL, Docker, Maven*
- Built a modular Java and TypeScript database tool supporting 10+ engines with schema-aware SQL generation, query validation, metadata exploration, error handling, automated testing, and Docker-backed CI/CD deployment.

PiXMod | Live | *Angular.js, Next.js, JavaScript, MongoDB, Stripe*
- Built a responsive full-stack product using Angular.js, Next.js, MongoDB, and Stripe, implementing maintainable user workflows, API integrations, automated validation, and subscription features that increased premium conversion 18%.

FinNexus | Live | *TypeScript, Next.js, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built a scalable platform supporting 50,000+ users and one million transactions, integrating secure REST APIs, real-time visualizations, MongoDB queries, Kubernetes deployment, production monitoring, and automated release workflows.